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Additional paleomagnetic results from the Sierra Nevada: Further constraints on Basin and Range extension and northward displacement in the western United States

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ABSTRACT

Modal analysis of new paleomagnetic results that have a skewed distribution from four plutons in the Kings Canyon National Park area, central Sierra Nevada, produces a pole position concordant with the results of Frei and others (1984) from the Yosemite area. These studies suggest (a) that stable magnetization resides in the more mafic plutons and inclusions in the Sierra Nevada and (b) that the amount of internal deformation within the central Sierra Nevada does not exceed a few degrees at the 95% confidence level.

The mean pole of both studies limits the relative displacement between the Sierra Nevada and cratonal North America to 6° ± 8° of clockwise rotation and 6° ± 6° of poleward displacement. The clockwise rotation agrees well with the model of Frei and others (1984) for Basin and Range extension, which predicts 2° of clockwise rotation for the Sierra Nevada. The poleward displacement suggests right-lateral shear of ~600 km within the Basin and Range between 90 and ~35 Ma.

INTRODUCTION

The major tectonic processes deforming the western margin of North America in the Tertiary are dextral shear associated with relative motion between the Kula, Farallon, and Pacific oceanic plates (Coney, 1978; Engebretson and others, 1985); the Laramide orogeny in the early Tertiary; and extension in the Basin and Range province. Basin and Range extension probably began in the Oligocene and has continued to the present (Lachenbruch and Sass, 1978; Zoback and Thompson, 1978; Eaton, 1979; Wernicke and others, 1982; Miller and others, 1983; and many others). The extent and interior position of the extending region with respect to the western margin of the North American plate led Hamilton and Myers (1966) to the conclusion that deformation of the western margin was a necessary consequence of the extension. Beck (1976, 1980) suggested that the clockwise rotations and the northward motion of crustal blocks in western North America, inferred from paleomagnetic studies, are the result of the dextral shear.

The location of the Sierra Nevada–Great Valley block between the Basin and Range province to the east and the region of complex accretionary geology to the west makes it especially interesting paleomagnetically. One possibility is that it shared the clockwise rotation and northward translation of the crustal blocks to the west. Alternatively, if it was simply the westernmost range of the Basin and Range province, it would have undergone several hundred kilometres of westward displacement during the Tertiary. If the amount of Basin and Range extension was greater in the north than in the south, the Sierra Nevada block must have undergone counter-clockwise rotation, and, conversely, it must have undergone clockwise rotation if the extension was greater in the south. Can this rotation be detected paleomagnetically?

In an earlier paleomagnetic study of Sierra Nevada plutons in the Yosemite area (Fig. 1), Frei and others (1984) were able to establish limits of 7° ± 11° for the amount of clockwise rotation since 90 Ma and 5° ± 8° for the amount of poleward translation. This result, however, was from one small part of an enormous plutonic complex. Moreover, the possibility of internal deformation has been raised by the report of Kanter and McWilliams (1982) of anomalous paleomagnetic results from the southern Sierra Nevada. Kanter and McWilliams (1982) demonstrated convincingly that the Tehachapi Mountains (Fig. 1) have, in fact, undergone 45° ± 14° of clockwise rotation since the Late Cretaceous and before 16 Ma. This raised the question of how much of the Sierra Nevada had been similarly deformed.

Lockwood and Moore (1979), studying microfaults and lineaments in the Sierra Nevada shown on aerial photographs (see Fig. 1), suggested that deformation by microfaulting reflects tectonic stresses related to regional extension in the Great Basin during the Cenozoic. Segall and Pollard (1981, 1983a, 1983b) mapped in detail fractures in the southern part of Mount Abbot quadrangle (Fig. 1) and found them to parallel the trends of the aerial photograph lineaments. In the western part of their study area, the fractures and lineaments trend north-northeast and were found to be joints and joint zones, respectively. In the central area, the lineaments and fractures trend northeast to east-northeast, and many of them are left-lateral faults and left-lateral fault zones, respectively. Similar conclusions were reached by Davies and Pollard (in press). Some internal deformation in the central Sierra Nevada is suggested by these studies, although displacements are small. Pervasive microfaulting was suggested as a possible explanation for the differences in paleomagnetic directions that were observed between two plutons studied in the Yosemite area (Frei and others, 1984).

A preliminary goal of the present study is to test paleomagnetically the extent of deformation within the Sierra Nevada. After assessing the effects of internal deformation, the regional tectonic implications of the paleomagnetic results will be considered.

GEOLOGIC SETTING

The Sierra Nevada batholith is composed predominantly of granitoids 220 to 80 Ma (Evernden and Kistler, 1970; Chen and Moore, 1982). The present study area in the region of Kings Canyon and Sequoia National Parks (Fig. 1) was chosen because radiometric ages (Chen and Moore, 1982) and geologic mapping (Ross, 1958; Moore and others, 1972; Moore, 1978; Sisson and others, 1984) were available. U-Pb ages of Lookout Peak pluton (Fig. 2a) suggest 97 Ma as the emplacement age (Chen and Moore, 1982). Field observations (Moore, 1978) and U-Pb age

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Figure 1. The location of paleomagnetic and deformational studies within the Sierra Nevada. Solid circle = area of present study in Marion Peak quadrangle (M) and Giant Forest quadrangle (G). T = Tehachapi Mountains area studied by Kanter and McWilliams (1982). A = Mount Abbot quadrangle, the locality of deformational studies by Segall and Pollard (1981, 1983a, 1983b) and paleomagnetic study by Ross (1984, personal commun.). Y = Yosemite area, Tower Peak quadrangle, the location of study of Frei and others (1984). Other quadrangles shown are those studied by Lockwood and Moore (1979). Inset: The geographical and tectonic setting of the Sierra Nevada (SN) and the Great Valley (GV) relative to the San Andreas (SA), Garlock (G), Walker Lane (WL), Brothers (B), Eugene-Denio (E), and McLaughlin (M) faults and fault zones; the Nevada Rift (NR); Basin and Range (BR); Colorado Plateau (CP); Cascades (C); Klamath Mountains (K); and Oregon Coast Range (CR).

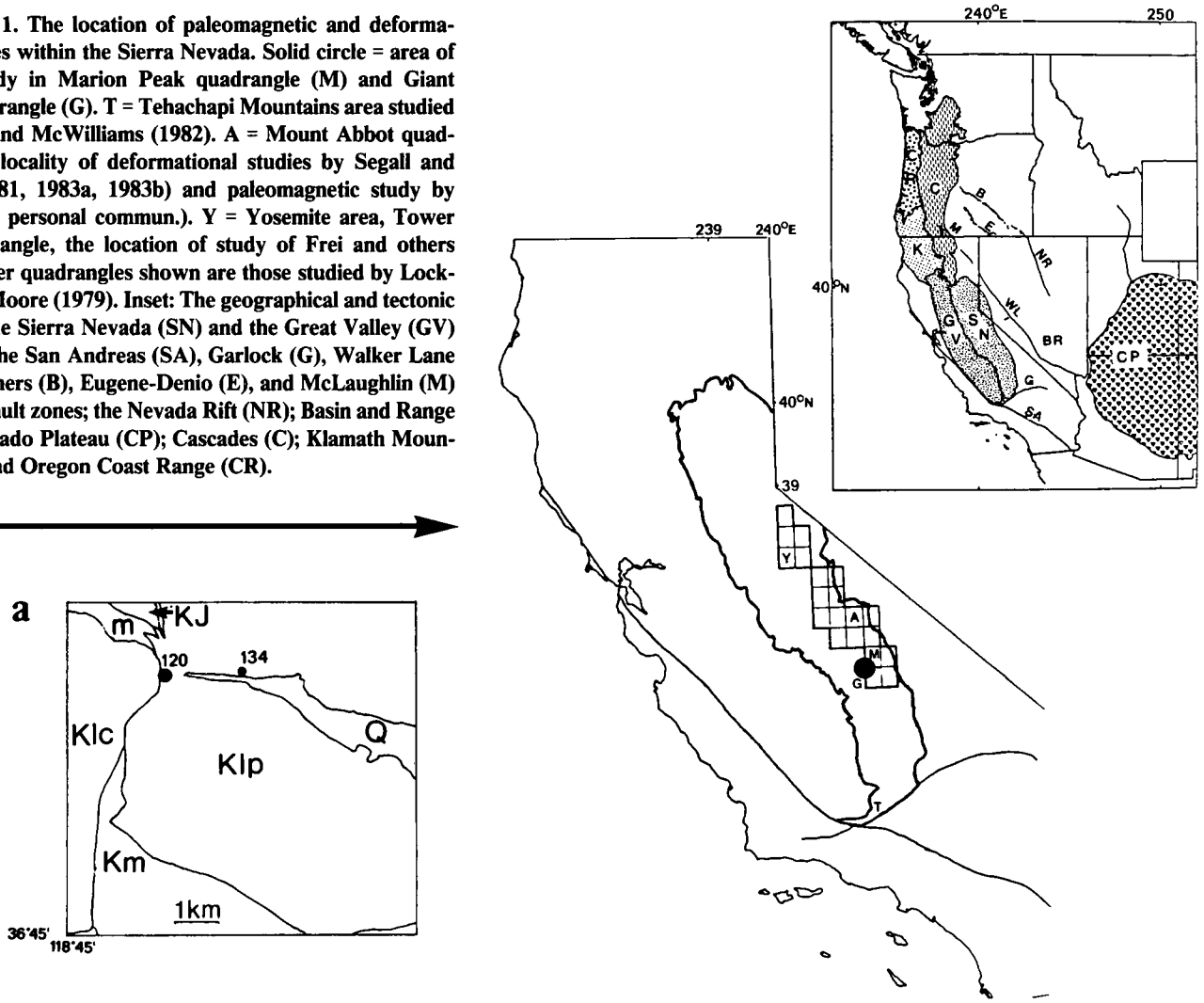


Figure 2. The sampling sites in Marion Peak quadrangle (above) and Giant Forest quadrangle (below). The site numbers are the same as those in Table 1. Square = the rejected site. Small circles = seven sites that show streaking directions. Large circles = nine sites that define the mode. (a) Southwest corner of Marion Peak quadrangle after Moore (1978). m = metamorphic rocks, KJ = sheared granite of Grand Dike, Klc = granodiorite of Lightning Creek, Klp = granodiorite of Lookout Peak, Km = granodiorite of Mitchell Peak, Q = Quaternary deposits. (b) Part of Giant Forest quadrangle after Ross (1958). m = metamorphic rocks, ggr = granite intrusive into Giant Forest granodiorite, gf = Giant Forest granodiorite, bm = Big Meadows quartz monzonite–granodiorite, wl = Weaver Lake quartz monzonite, a = alaskite, bgr = Big Baldy granite.

determinations (Chen and Moore, 1982) suggest that the Lookout Peak pluton intruded the Lightning Creek pluton. The U-Pb ages of the nested plutons of the Giant Forest–alaskite sequence (Fig. 2b) are 99 ± 3 Ma and indicate that the time of emplacement of the entire sequence is relatively short (1–2 m.y.) and that cooling of the plutonic complex from the margin to the center took ~3 m.y. The U-Pb ages of a mafic inclusion from this sequence are concordant and fall within the range of age of the host pluton (Chen and Moore, 1982). The observed intrusive relations

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within the sequence show that the granodiorite of Giant Forest is the oldest (U-Pb age of 97–102 Ma; Chen and Moore, 1982). It is intruded by the Big Meadows pluton (U-Pb age 98 Ma; Chen and Moore, 1982), which, in turn, is intruded by the Weaver Lake pluton (U-Pb age 97–99 Ma; Chen and Moore, 1982) (Ross, 1958; Sisson and Moore, 1984). Some samples to the south of the sequence give K-Ar ages 5–15 m.y. younger than the U-Pb ages of the Giant Forest–alaskite sequence. No anomalously young ages are reported, however, from the area which was sampled for the paleomagnetic study.

FIELD AND LABORATORY TECHNIQUES

A total of 129 cores from 17 sites were drilled in these 4 plutons (Fig. 2 and Table 1). From one to three specimens were prepared from each core. Pairs of specimens were selected from cores from 16 sites, one specimen for thermal and one for alternating magnetic fields (AF) demagnetization experiments. For the latter, the specimen was demagnetized using progressive alternating fields that had amplitudes ranging from 5 to 80 mT. The vector component diagrams, although noisy, suggest removal of a single component of magnetization with some randomly oriented magnetization being added during each demagnetization step. A second specimen from each of the 16 samples was stepwise thermally demagnetized using a Schonstedt TSD-1 shielded furnace. Noisy vector component diagrams were also observed for the thermally demagnetized specimens, suggesting removal of a single component of magnetization and the addition in each step of randomly oriented magnetizations. Most specimens exhibited a wide range of blocking temperatures with an obvious change of direction of magnetization in the range of ~550 to ~600 °C. The wide

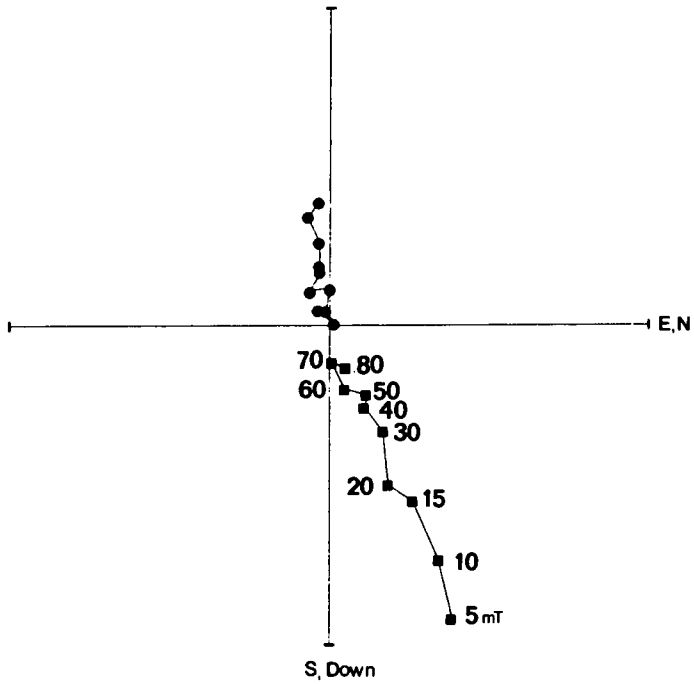


Figure 3. Vector component diagram of a sample from Giant Forest pluton during alternating field demagnetization. Circles = horizontal component. Squares = vertical component. The axis length is 1.0×10^{-6} Am^2.

TABLE 1. SITE MEAN DIRECTIONS

Site	In situ		Corrected		N	R	k	α_{95} or A_{95}	Site poles	
	Inc.	Dec.	Inc.	Dec.					N lat.	E long.
Big Meadows										
kc001	63.7	335.7	63.3	333.9	7	6.901	60.3	7.8	68.7	111.7
kc009	60.6	041.1	61.2	39.7	7	6.883	51.1	8.5	59.2	308.3†
kc026	69.1	33.2	69.6	30.9	8	7.104	7.8	21.2	62.9	283.5*
kc034	63.2	6.3	63.3	4.3	5	4.929	56.0	10.3	81.2	261.6†
kc061	53.3	358.2	53.3	356.9	6	5.934	75.6	7.8	86.2	103.6†
kc106	71.1	12.2	71.3	9.3	7	6.850	39.9	9.7	67.8	256.4†
kc113	60.1	336.4	59.7	334.8	6	5.951	101.0	6.7	70.0	170.0
Giant Forest										
kc019	65.6	343.7	65.3	341.6	7	6.943	105.8	5.9	72.8	135.2
kc041	63.0	327.0	62.4	325.4	6	5.905	52.5	9.3	62.9	176.9
kc056	63.9	337.9	63.5	336.0	5	4.982	219.1	5.2	70.1	183.5
kc069	56.1	310.4	55.3	309.5	7	6.879	49.7	8.7	49.8	165.5†
kc083	58.9	328.2	58.4	326.8	7	6.934	90.3	6.4	63.8	166.6
Inclusion in Giant Forest										
kc076	63.9	327.3	63.3	325.6	6	5.982	281.3	4.0	62.9	179.4
Weaver Lake										
kc049	59.5	32.1	60.0	30.6	6	5.934	76.0	7.7	65.9	311.6†
kc099	61.6	330.9	61.1	329.3	7	6.931	86.4	6.5	65.8	173.8
Lookout Peak										
kc120	54.8	339.2	54.4	337.9	5	4.880	33.3	13.5	72.0	51.9
kc134	53.1	352.3	53.0	351.0	7	6.976	251.1	3.8	82.0	30.2†
All sites	..	..	63.8	347.0	17	16.056	17.0	8.9	76.9	197.0
All sites but*	..	..	63.0	344.8	16	15.191	18.5	8.8	76.1	189.8
All sites but *,†	..	..	61.5	332.3	9	8.960	200.3	3.7	68.0	175.4
Mode	..	..	..	..	..	..	..	3.0§	67	175

Notes: in situ = field directions uncorrected for tilt. Corrected = field directions after correction for 1° of tilt to the west. Inc. = inclination. Dec. = declination. N = number of samples or sites averaged. R = vector sum. k = estimate of precision parameter. α_{95} = radius of 95% confidence in degrees for individual sites. A_{95} = radius of 95% confidence in degrees as calculated for mean pole positions. Site poles are corrected for tilt.

*Site with $\alpha_{95} > 20°$.

†Sites showing skewness.

§The 95% confidence is β_{95} as defined by Van Alstine (1979).

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range probably reflects variations in grain size. The characteristic directions determined by the 2 procedures were compared statistically and found to be the same at the 95% confidence level. Magnetic cleaning was, therefore, used for all of the samples.

PALEOMAGNETIC RESULTS

Approximately 85% of the samples were found upon demagnetization to have a stable magnetization which was consistent within site. A linear trend could be identified using a computer program modified after Kirschvink (1980) or graphically, although the vector component plots were quite noisy (Fig. 3). Some of the samples in five sites showed two characteristic components. The lower coercivity component, although consistent within site, was inconsistent between sites and will not be considered further. Approximately 15% of the specimens did not have a stable and consistent magnetization. Approximately half of these did not have an identifiable stable component of magnetization, and the rest had a natural remanence which diverged markedly in direction from those of the other samples from that site. Most of these samples appeared to have two characteristic components that could not be separated completely into lower and higher coercivity components.

In stepwise isothermal remanence (IRM) experiments, specimens typically acquired saturation IRM in fields of 0.2–0.5 T, indicating that magnetite is probably the main ferromagnetic mineral present. The blocking temperatures that were described earlier also suggest that magnetite is the main remanence carrier.

The polarities of all specimens were normal. The observed U-Pb ages in the range 102–97 Ma suggest that the plutons cooled during the long Cretaceous normal period which lasted from 118 to 83 Ma (time scale of Harland and others, 1982).

ANALYSIS OF DATA

A characteristic direction was determined for each specimen using stepwise AF demagnetization (Fig. 3). These directions were combined using Fisher statistics to form site means (Table 1) and the corresponding pole positions (Fig. 4a). The pole positions in Table 1 and Figure 4a are corrected for 1° of tilt to the west of the Sierra Nevada (see Frei and others, 1984, for discussion of structural tilt correction). Mean values were calculated for several subsets of the original 17 sites using selection criteria described in the notes to Table 1.

A contour plot of the density of pole positions shows a markedly skewed distribution which has a high and circular concentration of nine site pole positions (Fig. 4b). If one assumes that the most reliable information resides in these nine sites, the best measure of central tendency for these measurements is provided by the mean of these nine poles (Table 1). The locations of the nine sites included in this mean are marked by the large circles in Figure 2. Their distribution among the sites that have more scattered poles rules out faulting or tilting as the reason for the difference between the two sets of sites. Secular variation is averaged at the intra-site level in plutons this large (Frei and others, 1984) and, therefore, is not a likely cause for the dispersion of the more scattered pole positions. Part of the skewness consists of streaking toward the present geographic pole and thus appears to be an overprint parallel to the present axial dipole field. The origin of the remainder of the skewness is uncertain.

Modal analysis is more advantageous than are Fisher statistics (Fisher, 1953) for skewed distributions, as Fisher statistics apply to distributions that are symmetrical about the mean. The mode was calculated using a program, modified after Van Alstine (1979, 1980), using a smoothing parameter of 1.0, a grid spacing of 1.0, and 9 points near the mode. The mode and the mean of 9 sites are only 1° apart. The mean pole will be used in subsequent discussions.

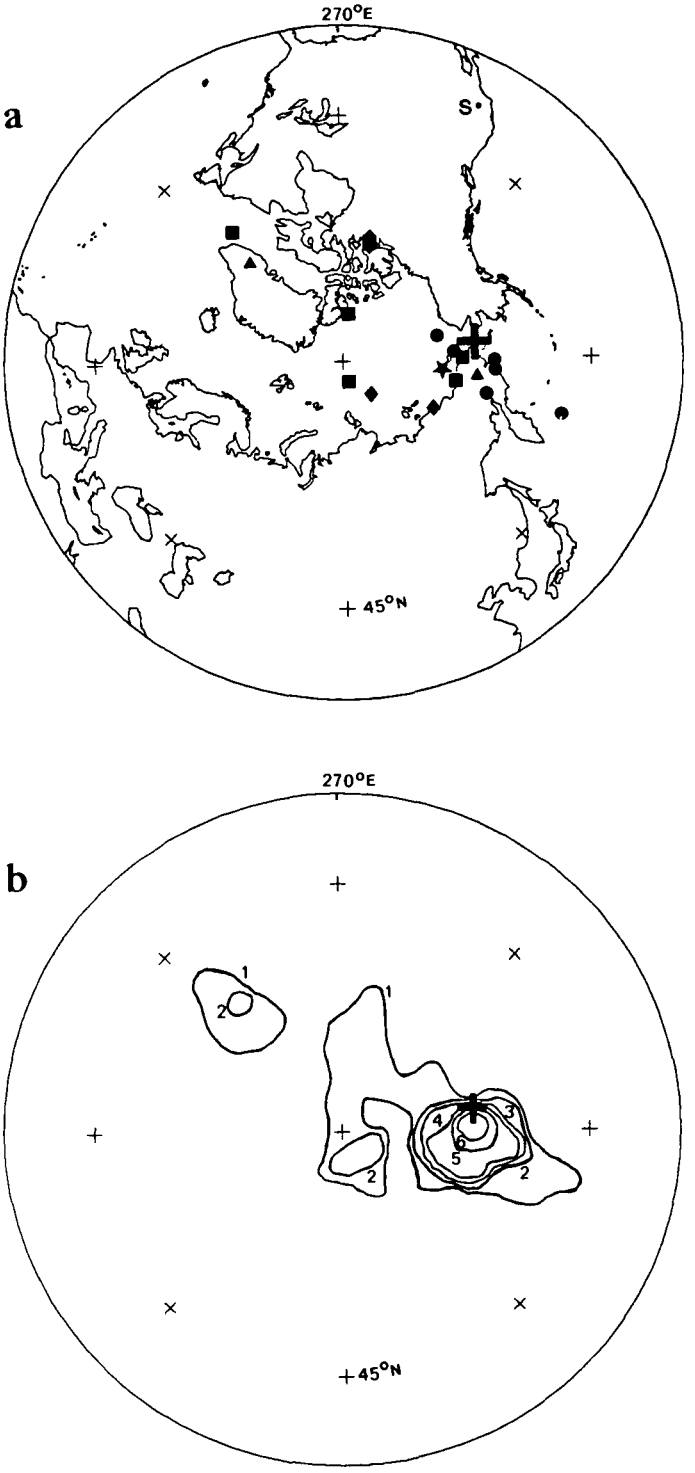


Figure 4. (a) The 16 site pole positions grouped by pluton. Circles = sites from Giant Forest granodiorite. Squares = sites from Big Meadows quartz monzonite. Triangles = sites from Weaver Lake quartz monzonite. Diamonds = sites from Lookout Peak granodiorite. Square with triangles = mixed site of Big Meadows quartz monzonite and Weaver Lake quartz monzonite. Star = mean of the results from the Yosemite area (Frei and others, 1984). Large plus symbol = cratonal mean pole (see text). S = site location. (b) Contours of density of site poles. The numbers outside each contour refer to the density of points within the contour. Large plus symbol = cratonal mean pole.

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Figure 5. Site pole positions for the data from the Yosemite area (Frei and others, 1984). (a) The site pole positions from the three plutons. Triangles = results from inclusions within Grace Meadows. Squares = fine-grained Mount Gibson. Circles = coarse-grained Mount Gibson. The large plus symbol is the expected cratonal pole (see text). (b) Contours of the density of the 40 site pole positions shown in part a. The numbers outside each contour refer to the density of points within the contour. Contours 5, 7, 11, and 13 omitted for clarity.

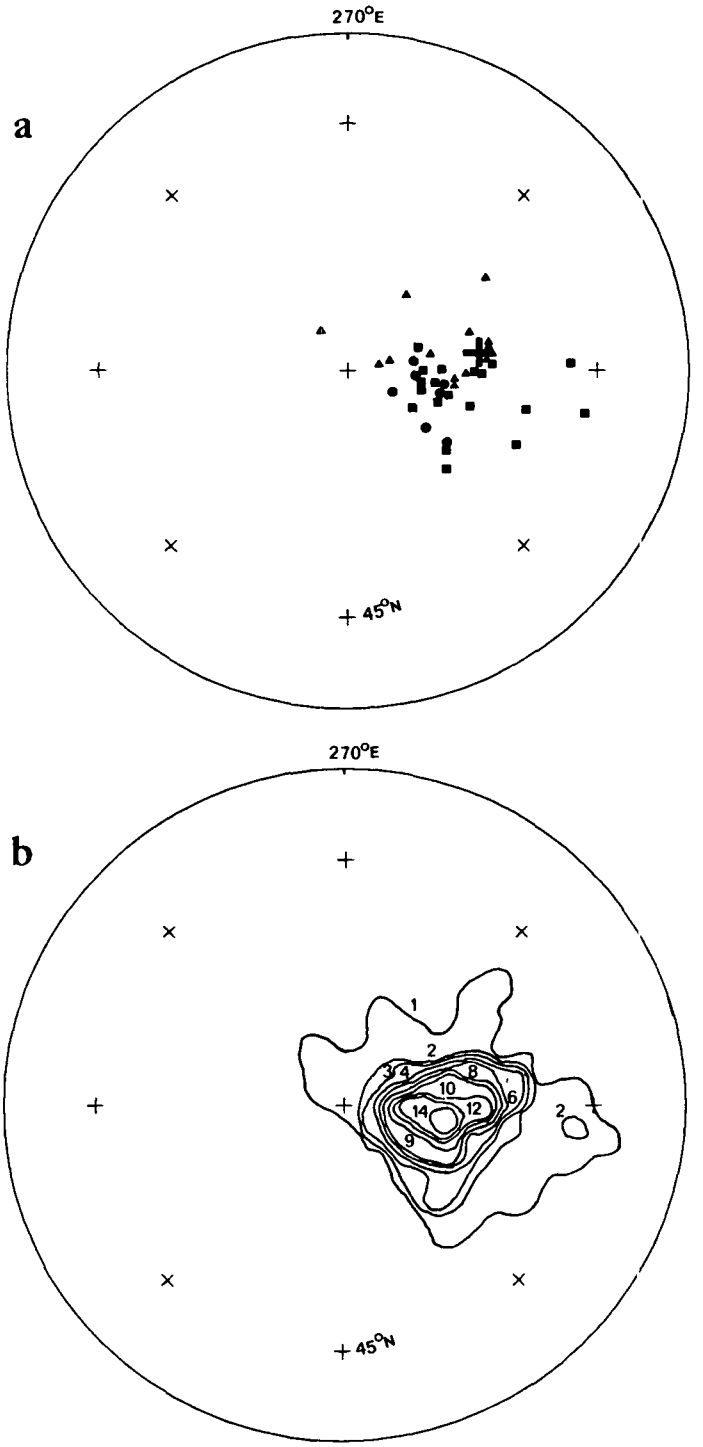


TABLE 2. PALEOMAGNETIC RESULTS FROM THE SIERRA NEVADA

Rock type or locality	Pole lat. degrees N	Pole long. degrees E	N	A_{95}/β_{95}	Reference
Reeve Formation	72.5	141.8	5	19.3	1
Metamorphic belt	64.5	165.2	3*	8.4	2
Bucks batholith	57.6	194.8	9	7.9	3
Cretaceous granites	69.0	195.0	14	9.6	4
Yosemite plutons	72.6	174.7	3	9.4	5
This study (a)	76.1	189.8	16	8.8	6
This study (b)	68.0	175.4	9	3.7	6
This study mode	67	175	16	3	6
Sierra Nevada					
i	73.6	177.9	4	6.5	5, 6
ii	72.9	178.8	56	3.3	5, 6
iii	71.5	174.9	4	6.3	5, 6
iv	71.2	174.8	4	6.5	5, 6
v	70.9	175.6	49	2.6	5, 6
vi	71.7	176.4	54	2.8	5, 6

Notes: References are as follows. 1 = Hannah and Verosub, 1979; 2 = Bogen and others, 1985; 3 = Grommé and others, 1967; 4 = Grommé and Merril , 1965; 5 = Frei and others, 1984; 6 = this study. As to number of sites, a includes 16 sites; b 9 sites (see Table 1 and text).

*N in this study is number of formations. It includes 10 sites. i = mean of 3 plutons from reference 5 and mean a from this study. ii = mean of 56 sites, 40 from reference 5 and 16 from this study. iii = mean of 3 plutons from reference 5 and mean b from this study. iv = mean of 3 plutons from reference 5 and mode from this study. v = mean of 49 sites, 40 from reference 5 and 9 from this study. vi = mean of 54 sites, 40 from reference 5 and 14 from this study [excluding sites kc009 and kc049, the only sites that suggest skewness if all 56 sites are plotted (Fig. 6a)].

DISCUSSION

Before considering some possible tectonic interpretations of these results, two preliminary issues must be addressed. First, is the measured magnetization a stable remanence acquired when the plutons cooled? Second, has the Sierra Nevada undergone large-scale internal deformation?

Stability of Magnetization in Plutonic Rocks of the Sierra Nevada

In the case of Cretaceous plutons, magnetic stability cannot be investigated using fold tests, conglomerate tests, or antipodal normal and reversed direction. The case for the stability of the magnetization rests mainly, therefore, on laboratory experiments and internal consistency over a broad sampling area. The nine sites that have pole positions contributing to the third mean in Table 1 include five of the six sites from the Giant Forest pluton and its inclusion, two of the five sites from the Big Meadows pluton, and one site from each of the two sites in the Weaver Lake and Lightning Peak plutons. The site which includes samples from both Big Meadows and Weaver Lake was not included in the mean. These results show that the sites from the Giant Forest pluton had the smallest scatter. Because grain size is similar in the Big Meadows and Giant Forest plutons, the higher scatter of the site pole positions from the Big Meadows pluton could be a result of different mineralogical compositions. The two Mount Gibson plutons studied by Frei and others (1984), which showed very stable magnetizations and low scatter, have a higher fraction of mafic minerals than do the plutons studied here. The Giant Forest pluton has, on the average, a higher percentage of mafic minerals (biotite and hornblende) than does the Big Meadows pluton. No difference in magnetic blocking temperature, in IRM saturation, or in natural remanent magnetization (NRM) intensity, however, could be seen between the two plutons.

The few samples that were collected from mafic inclusions within the plutons in this study and the samples collected from the inclusions in the Grace Meadow pluton in the Yosemite area (Frei and others, 1984) showed stable magnetizations and low scatter. Although the inclusions are finer grained than are the plutons, in the Sequoia sequence they were either derived from the same source as the granodiorites or totally equilibrated with the granodioritic magma (Chen and Moore, 1982). Grommé and Merrill (1965) also observed higher stability of magnetization in inclusions than in host granodiorites in the Sierra Nevada. It appears that the more mafic and finer grained the pluton, the higher the stability of magnetiza-

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tion. The case for the paleomagnetic validity of the results from the Sierra Nevada rests on the internal consistency of results from the finer grained, more mafic lithologies at sampling sites spanning 150 km.

Constraints on Internal Deformation in the Central Sierra Nevada

In the Yosemite area, a small (8°) but statistically significant difference in paleomagnetic directions could be seen between the highly sampled Grace Meadow pluton and the two Mount Gibson plutons (Frei and others, 1984), due possibly to microfaulting as described by Lockwood and Moore (1979). The present study at sites 150 km to the south of Yosemite, together with results from other studies, provides an opportunity to assess the amount of deformation over distances of hundreds of kilometres.

Table 2 is a summary of available paleomagnetic data from the central Sierra Nevada. The results of Kanter and McWilliams (1982) from the southern Sierra Nevada that show significant rotations are not included in the table. The results from the Permian Reeve formation reported by Hannah and Verosub (1980) fail the fold test. Hannah and Verosub (1980) suggested that the remagnetization is Late Jurassic because of the location of this pole relative to the Cratonal North American Polar Wander Path. Using this result to constrain deformation is a circular argument, and therefore this pole will not be used. The Logtown Ridge and Penon Blanco Formations and the Sonora dike swarm from the Sierra Nevada foothills are Jurassic in age and have all been metamorphosed (Bogen and others, 1985). Two components of magnetization were found. The northwestern component is secondary, interpreted by Bogen and others (1985) to be related to the Nevadan metamorphism (late Kimmeridgian). The age of this component of magnetization is not well established, however, and could be Late Jurassic or Cretaceous. Bogen and others (1985) suggested that the component of magnetization directed to the southeast is primary, of Late Jurassic age. This component of magnetization was found in three sites within the Logtown Ridge Formation and in four dikes of the Sonora dike swarm. No tilt correction is available for these dikes, and four dikes are not enough to average out secular variation. The Bucks batholith (Grommé and others, 1967) is Early Cretaceous in age, is located west of the Melones fault zone, and might have been tilted during the intrusion of the younger plutons. Most samples in the study of Upper Cretaceous granites (Grommé and Merrill, 1965) were not demagnetized, and, therefore, care should be taken in using this result. The two studies of Late Cretaceous Sierra Nevada plutons wherein modern cleaning techniques were used are the present study and the earlier study from the Yosemite area. Frei and others (1984) reported two mean poles from the Yosemite area, one the mean of the three plutons, resulting in the pole at 72.6°N, 174.7°E, with an A_{95} of 9.4°, and the other calculated using the results from the 40 sites (71.6°N, 175.6°E, A_{95} of 3.1°). The poles are close, and the larger A_{95} is probably the more realistic. Figure 5 is a plot of the distribution of the pole positions of the 40 sites from the 3 plutons in the Yosemite area (Frei and others, 1984) and the contours of their concentration. The skewness that is characteristic of the present study is absent from the Yosemite data.

In order to determine the total internal deformation which could have taken place in the ~150 km between the two areas studied, the mean pole positions of the two studies were compared. Using result b in Table 2, the relative tilt of the southern site relative to the northern site is 0.8° ± 5.5° (95% confidence) and the rotation is −6.1° ± 10.6°. [The equations used for calculating the tilt, the rotation, and their 95% confidence limits are after Beck (1980), Cox (1980), and Demarest (1983)]. Neither the relative tilt nor the relative rotation are significant. In order to assess the sensitivity of this result to assumptions made in the reduction of the data, a similar calculation was done using mean pole a in Table 2. The values for the relative tilt and rotation were 2.3° ± 8.4° and 6.3° ± 13.8°, respectively.

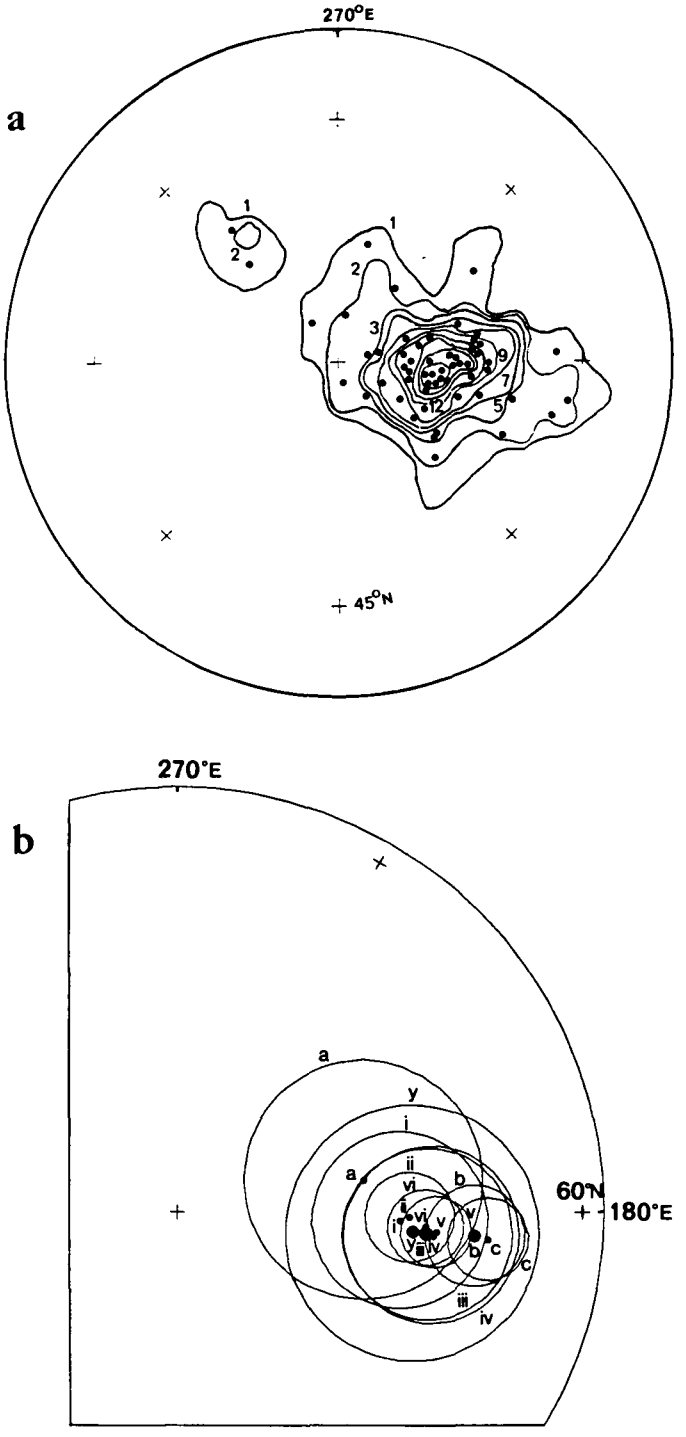


Figure 6. (a) Site pole positions (56) of sites from Yosemite (Frei and others, 1984) and this study with density contours. The numbers outside or on each contour refer to the density of points within the contour. The three most inner contours (14, 16, and 20) are not numbered. Contours 4, 6, 8, 10, 11, 13, 15, and 17–19 were omitted for clarity. (b) Mean pole position from Sierra Nevada studies and ovals of 95% confidence. a and b = means from Table 2. c = mode from Table 2 with confidence ovals β_{95} as defined by Van Alstine (1979, 1980). y = mean of the study in the Yosemite area (Frei and others, 1984). i–v = different means of the two studies as in Table 2. The means b, y, and iii are marked with bigger circles as they are taken as means for the two studies and for the Sierra Nevada (see text).

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Again, the values are insignificant at the 95% confidence level. These results indicate that relative rotations larger than ~10° are not present in the central Sierra Nevada. The large rotation in the southern Sierra Nevada described by Kanter and McWilliams (1982) does not extend as far north as the present study area.

The Relative Displacement Between the Sierra Nevada and Cratonal North America

A comparison of the mean pole of the Sierra Nevada and a mean cratonal pole of the same age shows the amount of relative rotation and poleward motion between the two. The data from Yosemite and the present study are grouped in six different ways (i–vi) in Table 2 and Figure 6b. All of the means and modes are close to each other and not significantly different at the 95% confidence level. For reasons discussed by Frei and others (1984), the poles that were calculated assuming that each site is independent have confidence limits that are too small. In subsequent analysis, we will take the extremely conservative view that only four poles are independent, the three from the plutons in the Yosemite area and pole b of this study (Table 2). The mean of these four poles (entry iii in Table 2) is at 71.5°N, 174.9°E, with N = 4 and A_{95} = 6.3°. It is this pole which will be used to constrain displacement between the Sierra Nevada and cratonal North America.

In order to determine relative motions between the Sierra Nevada as a whole and cratonal North America, a reference pole for the craton is needed. Frei and others (1984) compared their results from the Yosemite area to three reference poles, each calculated in a slightly different way, and preferred the pole of Irving and Irving (1982). Because of the importance of the cratonal pole, for the present analysis we have reviewed all published results from rock units that have ages ranging from 110 to 80 Ma from cratonal North America. The following studies were included in our final analysis: (a) Niobrara Formation (Shive and Frerichs, 1974), as recalculated for the lower part of the sequence by Gordon and others (1984); (b) alkalic rocks from Arkansas (Scharon and Hsu, 1969); and (c) Isachsen diabase (Larochelle and Black, 1963, and Larochelle and others, 1965), as recalculated by Gordon and others (1984). The other available studies were excluded for one or more of the following reasons: age of magnetization not well determined; possibility of an overprint; insufficient cleaning, although results suggest that cleaning is necessary; structural control not well defined; sampling site not part of the craton. The three poles are closely grouped (Table 3 and Fig. 7), whereas localities are far apart, an encouraging indication of consistency. Due, in part, to the fact that the North American pole was moving very slowly at this time, the mean, at 66.4°N and 187.4°E (A_{95} is 3.6°), will be used as the cratonal reference pole. This pole is very similar to the poles of Irving and Irving (1982) for 90 and 100 Ma. The somewhat different poles calculated by Harrison and Lindh (1982) and by Van Alstine and de Boer (1978) use the pole determined by Grommé and Merrill (1965) for plutonic rocks from the Sierra Nevada and, therefore, cannot be used to evaluate displacement between the Sierra Nevada and the craton.

This cratonal pole and our new Sierra Nevada pole give a poleward displacement, *d*, of the Sierra Nevada relative to the craton of 5.5° ± 5.6° (95% confidence) and a clockwise rotation, *r*, of 5.7° ± 7.9° (see Frei and others, 1984, for equations used). These results, although similar to the ones determined by Frei and others (1984), have smaller error bars as a result of more accurate cratonal and Sierra Nevada poles.

The positive value found for the poleward displacement is consistent with distributed right-lateral shear in western North America and with the results of Kanter and McWilliams (1982) for the Upper Cretaceous rocks from the southern Sierra Nevada. Larger poleward displacements have been reported from mid-Cretaceous plutons in British Columbia (Beck and others, 1981), in Baja California (Hagstrum, 1985), and from the Salinian block (Kanter, 1983; Champion and others, 1984). The results from the

TABLE 3. NORTH AMERICAN CRATONAL POLES FOR THE PERIOD ~80–~110 Ma

Formation	Site location N lat.	Site location E long.	Age	Pole position N lat.	Pole position E long.	σ or A_{95}	Ref.*
Niobrara	38.5	257.6	85–89	66	192	5	1, 2
Alkalic rocks	34.5	267.2	98 ± 5	65	187	9	3
Isachsen diabase	78.8	256.2	109 ± 5	68	183	11	4, 5, 2
Mean	..	..	..	66.4	187.4	3.6	

Notes: σ = the standard error as calculated by Gordon and others (1984) for the three studies. A_{95} = the 95% confidence limit on the mean pole. The time scale used is that of Harland and others (1982).

*The references are as follows. 1 = Shive and Frerichs (1974), 2 = Gordon and others (1984), 3 = Scharon and Hsu (1969), 4 = Larochelle and Black (1963), 5 = Larochelle and others (1965).

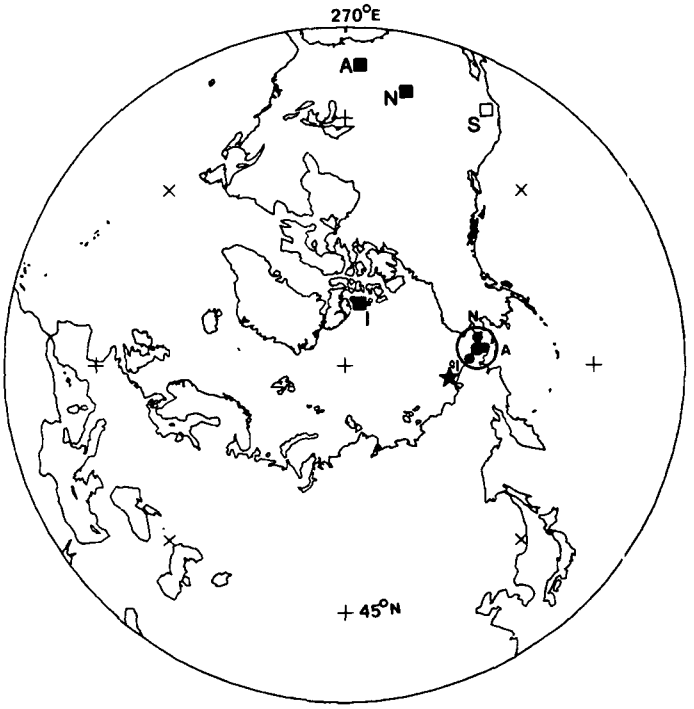


Figure 7. North American cratonal poles of ages ~80–~110 Ma (circles) and their sampling localities (squares) (see Table 3). A = Alkalic rocks, N = Niobrara formation, I = Isachsen diabase. S = present study area. Star = pole iii in Table 2 of the Sierra Nevada. The mean of the three cratonal poles and its oval of 95% confidence are shown.

Miocene rocks from the southern Sierra Nevada (Kanter and McWilliams, 1982) suggest that the poleward displacement is pre–16 Ma. The paleomagnetic results from the Cascades (Magill and Cox, 1980) and the overlap relations of the Cascades, Klamaths, Great Valley, and Sierra Nevada suggest that most of the displacement is pre–35 Ma. In summary, the paleomagnetic data from the Sierra Nevada are consistent with from 0 to 1,300 km of poleward displacements sometime between 90 and ~35 Ma. If the displacement is as large as the mean value of ~600 km, it seems likely that it is distributed across the Basin and Range province as right-lateral shear predating high-angle normal faulting.

In the following discussion, the implications of the paleomagnetic results for the reconstruction of Basin and Range extension will be considered. If, for example, the paleomagnetic data suggest that X° of clockwise rotation took place since some time in the past, then a counter-clockwise rotation of X° (clockwise rotation of –X°) will be used in the reconstruction to that same time.

Because the observed possible poleward displacement would have taken place before Basin and Range extension, we will assume that the Sierra Nevada was almost at its present latitude by 30 Ma. Displacement

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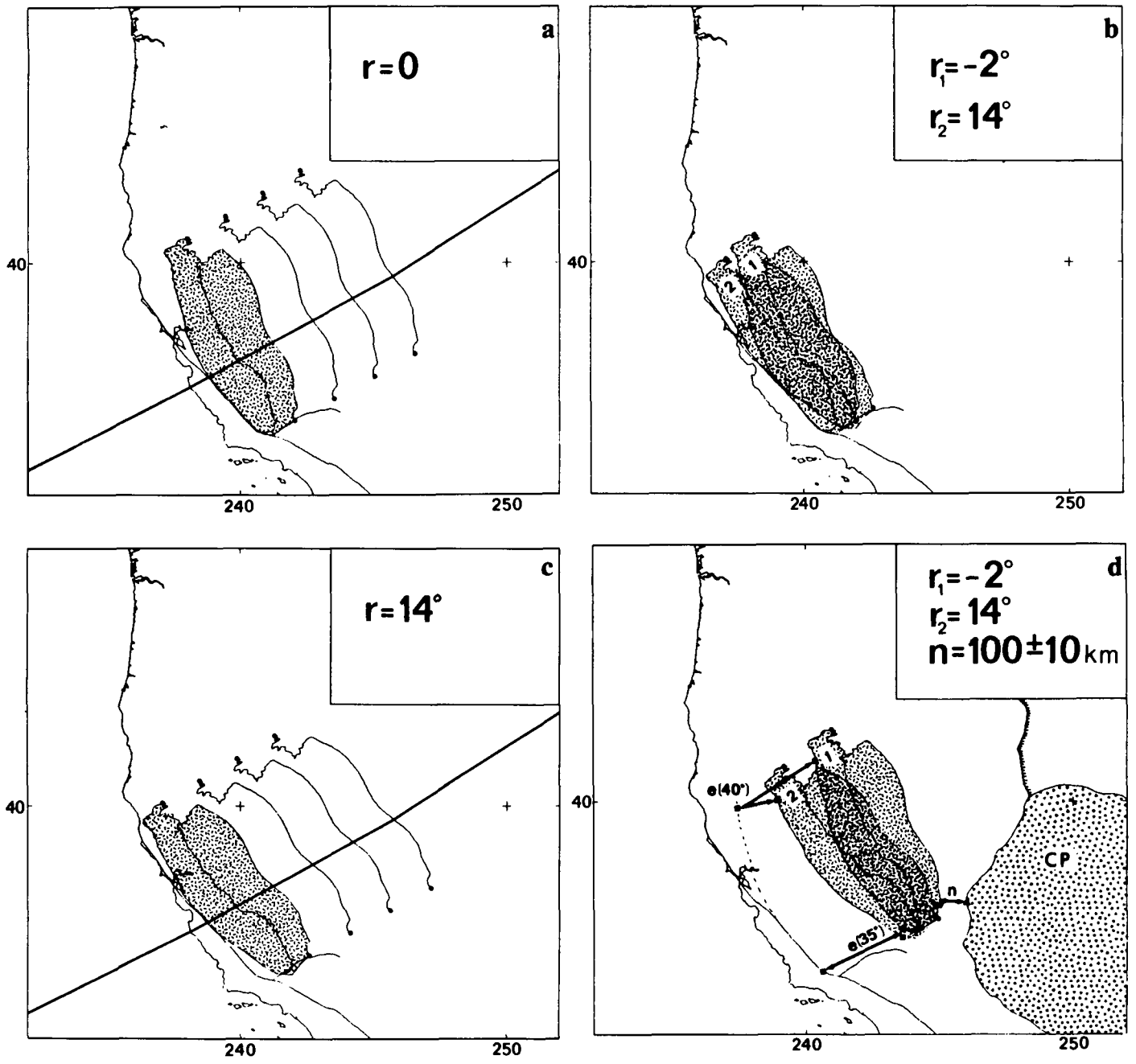


Figure 8. Limits on reconstruction of Basin and Range extension established by paleomagnetic results from the Sierra Nevada and the position of the Colorado Plateau. r is the paleomagnetically determined amount of clockwise rotation undergone by the Sierra Nevada since it formed. (a) Possible reconstructions of Sierra Nevada block for r = 0 ignoring constraints imposed by location of Colorado Plateau. (b) Limits of clockwise (1) and counter-clockwise (2) rotations for reconstructions of the Sierra Nevada block consistent with the paleomagnetically determined rotation r = 6° ± 8°. (c) Some possible reconstructions of the Sierra Nevada block (r = 14°) ignoring constraints imposed by the Colorado Plateau. (d) Reconstructions of the Sierra Nevada block consistent with paleomagnetically determined rotations (r = 6° ± 8°) and with pinning of the southeast corner of the Sierra Nevada block at a distance n = 100 ± 10 km from the Colorado Plateau (CP) which, together with hachured line to the north, mark the eastern boundary of the Basin and Range province. e (40°) arrows show displacement of the Sierra Nevada block at latitude 40°N for r = -2° and r = 14°, and e (35°) arrow shows displacement at 35°N, which is not sensitive to the angle of rotation. The total rotation poles producing each model are given in Table 4.

along a small circle about the cratonal paleomagnetic pole is demonstrated in Figure 8a. The four positions shown by the outlines of the eastern boundary of the Sierra Nevada block all have exactly the same paleomagnetic pole, and thus this motion, which can result in large amounts of extension, could not be detected paleomagnetically. The small amount of possible rotation of the Sierra Nevada block serves to constrain how much Basin and Range extension may have varied from north to south. As one extreme example, if the southeast corner of the Sierra Nevada block had remained pinned against the Colorado Plateau and extension had increased northward by rotation of the Sierra Nevada block about this pivot point, we would observe large counter-clockwise rotation of the Sierra Nevada block. This can be ruled out, as can pinning of the northeast

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corner of the Sierra Nevada block. Counter-clockwise rotations about a local axis of –2° and 14° corresponding to the 95% confidence limits of the paleomagnetic results have been used in Figure 8b to reconstruct the positions of the Sierra Nevada. The reconstructions in Figures 8a and 8b are consistent with paleomagnetic data but not with geologic observations to the east.

Combinations of these two displacements are possible. In adding rotations, the sequence of addition is important if the rotations are large. In these reconstructions, the local rotations are performed first and then the rotation about the expected cratonal paleomagnetic pole is performed. The resultant rotations are listed in Table 4. Figure 8c demonstrates different amounts of rotation (0°, 2°, 4°, and 6°) about the expected paleomagnetic pole added to the rotation about the local pole of Figure 8b(2). The range of models consistent with the paleomagnetic data is endless until constraints are introduced based on the location of the Colorado Plateau to the east. Figure 8d shows 2 reconstructions for local rotations of –2° and 14° with 4° and 3°, respectively, of rotation about the expected paleomagnetic pole (66.4°N, 187.4°E). Both are consistent with the paleomagnetic results. Also shown in Figure 8d is the geologic constraint imposed by the location of the Colorado Plateau. Larger values of rotation about the cratonal pole would reduce the distance between the southeastern end of the Sierra Nevada and the Colorado Plateau (n in Fig. 8d). Because basement rocks are exposed between the two, the distance n between the Sierra Nevada and the Colorado Plateau must be greater than zero in

TABLE 4. FINITE ROTATION POLES USED TO RECONSTRUCT THE SIERRA NEVADA FROM PRESENT POSITIONS TO POSITION AT 30 Ma

Reconstruction in figures 8 and 9	Finite rotation pole		
	N lat.	E long.	CCW rotation*
8a(1)	66.4	187.4	..
8a(2)	66.4	187.4	2.0
8a(3)	66.4	187.4	4.0
8a(4)	66.4	187.4	6.0
8b(1)	37.4	240.7	–2.0
8b(2)	37.4	240.7	14.0
8c(1)	37.4	240.7	14.0
8c(2)	41.94	238.24	15.55
8c(3)	45.6	235.96	17.19
8c(4)	48.58	233.85	18.9
8d(1)	58.99	125.7	2.84
8d(2)	43.87	237.08	16.38
9a	58.99	125.7	2.84
9b	65.87	225.82	4.47

*Counter-clockwise rotation in reconstruction.

reconstructions. The clockwise rotation described by Kanter and McWilliams (1982) is reported only from locations to the southwest of the point closest to the Colorado Plateau in the reconstruction (Fig. 8d). If this deformation also took place as far north as this point, additional space would have to be left between the Sierra Nevada and the Colorado Plateau in the reconstructions for this part of the Sierra Nevada. If the southeast corner of the Sierra Nevada block was ~100 km from the Colorado Plateau, then the two extreme possible positions of the block prior to Basin

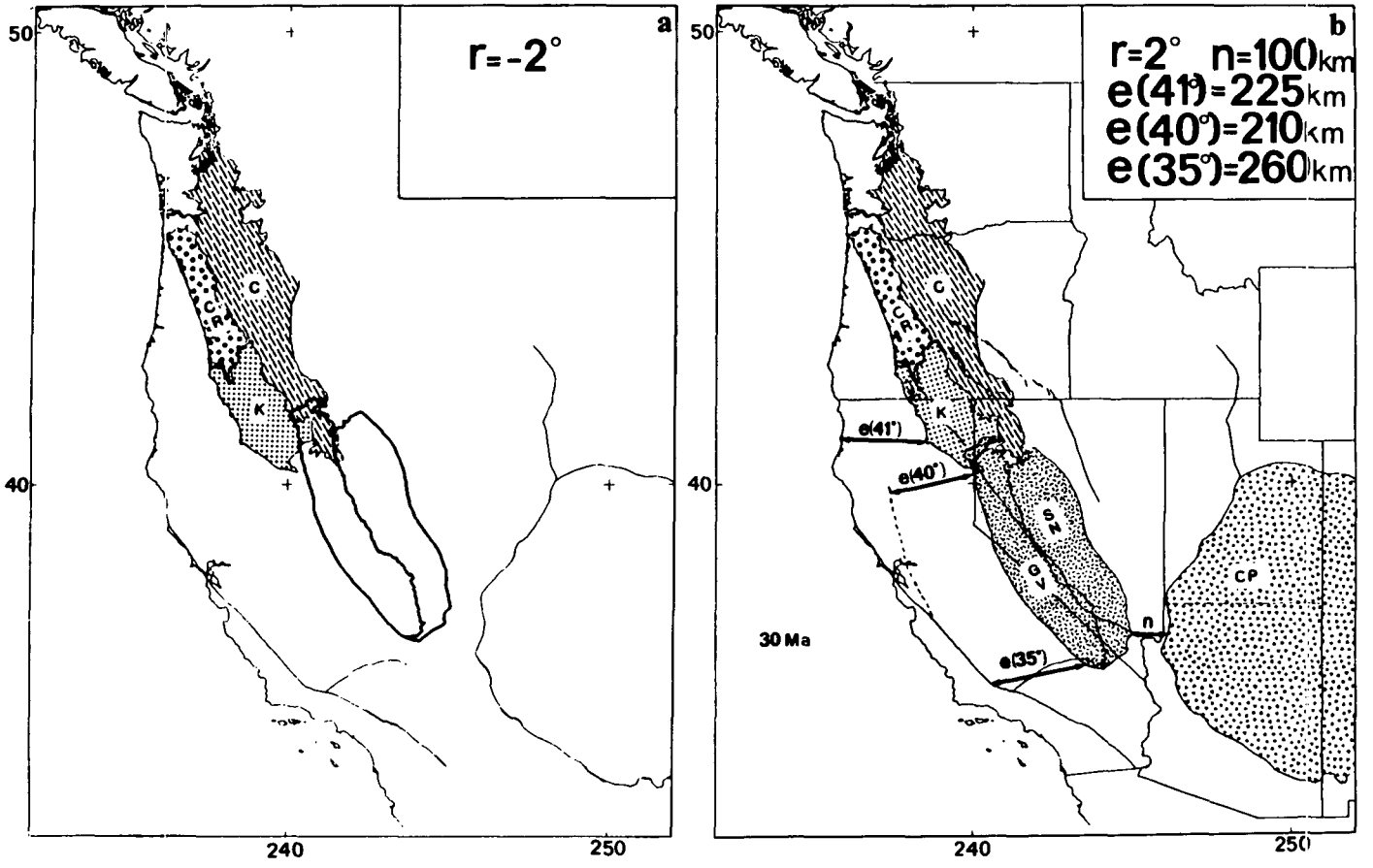


Figure 9. Reconstructions of the Sierra Nevada block superimposed upon reconstruction of crustal blocks to the north. K = Klamath Mountains. CR = Oregon Coast Range. C = Cascade Mountains (after Frei and others, 1984). (a) Reconstruction of the Sierra Nevada block as in Figure 8d(1) (r = –2°) showing unacceptable overlap with the Klamath Mountains and the Cascade blocks. (b) Reconstruction which best satisfies paleomagnetic and geologic constraints.

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and Range extension, consistent with the paleomagnetic data, would be as shown by the stippled pattern. The amounts of Basin and Range extension at latitudes 35°N and 40°N are marked in Figure 8d as e (35°) and e (40°). If the westward translation, e (35°), at the southern end of the Sierra Nevada (latitude 35°N) was ~300 km, then the translation e (40°) at the north end (latitude 40°N) is constrained by the paleomagnetic rotations to the range 130 to 290 km (Fig. 8d). Greater amounts of extension at the north end of the Sierra Nevada would have produced clockwise rotation greater than −2°, outside the 95% confidence limits provided by the paleomagnetic results.

The reconstruction of the Sierra Nevada in Figure 9a is the same as that in Figure 8d(1). A second geologic constraint, however, is introduced. This is the reconstructed position of the Oregon Coast Range, the Klamath Mountains, and the Cascades. The overlap relations between the Hornbrook Formation, which is part of the Great Valley sequence in the Klamath Mountains area, and the Klamath Mountains and between the Cascades and the Klamaths were discussed by Magill and Cox (1980) and by Frei and others (1984). The relative positions of the Sierra Nevada and the Klamaths and Cascades as reconstructed in Figure 9a are in disagreement with these constraints, suggesting that some (~1°) poleward motion took place during Basin and Range extension. In the preferred reconstruction (Fig. 9b), the clockwise rotation of the Sierra Nevada is 2°. The values for Basin and Range extension are ~220 km at 40°N, directed east-west, and ~260 km at 35°N, directed west-southwest, and the distance between the Colorado Plateau and the southern end of the Sierra Nevada is ~100 km.

CONCLUSIONS

Internal deformation within the central Sierra Nevada is no greater than ~10° between structurally simple regions ~150 km apart. Whether this will also be true in local regions where substantial amounts of internal deformation are suggested by microfaults awaits further study. The mean 100–90 Ma pole of the Sierra Nevada is at 71.5°N and 174.9°E with an A_{95} = 6.3°. This result is consistent with a tectonic model beginning with a modest amount (~600 km) of poleward displacement of the Sierra Nevada due to right-lateral shear in the Basin and Range sometime between Late Cretaceous and Miocene followed by Basin and Range extension of nearly equal magnitude between latitudes 35°N and 40°N.

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